YIFEI SUN

Principal Software Engineer · Backend & Platform Architecture · 13 Years

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SUMMARY

Distinguished-level backend & platform engineer with 13 years designing and operating large-scale distributed systems, setting long-term technical direction, and driving platform-wide architectural standards. Experience architecting highly available systems (99.9% availability), building internal developer platforms and shared services adopted across organizations, and leading complex cross-cutting initiatives spanning multiple teams and departments. Track record of identifying and resolving platform-wide tech debt and architectural bottlenecks, influencing engineering-wide practices in observability, reliability, and performance, and mentoring senior engineers toward Principal-level scope.

LEADERSHIP HIGHLIGHTS

- Led cross-cutting platform initiatives involving 10+ engineering teams across 3 departments
- Built internal developer platforms, shared services, and observability practices adopted broadly across the organization
- Set long-term technical direction for API architecture, platform services, and developer experience via RFCs and design docs
- Mentored senior engineers to Principal-level scope; authored design docs and architecture standards

Technologies: Java, Spring Boot, GraphQL (Netflix DGS), Python, Node.js · AWS, Kubernetes, Kafka, Redis, PostgreSQL · Platform Architecture, Microservices, Event-Driven, CQRS · Datadog, Grafana · Keycloak, SSO, OAuth 2.0

INDEPENDENT PROJECT

Configurable Platform Engine & Experimentation System

- Architected a hybrid rules + multi-path inference engine with a policy overlay over heterogeneous data sources, producing explainable real-time decisions with full observability and auditability.
- Designed a configurable policy framework with DB-backed enterprise configs, environment-specific overrides, and a scheduler-driven tuning loop that sweeps parameters and publishes promotable config versions — enabling safe, iterative platform improvement.
- Built an experimentation and evaluation system for end-to-end simulation, offline backtesting, and parameter search (grid & Bayesian optimization) with robust scheduling, observability, and multi-provider support.

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead

Oct 2019 – Present

GraphQL Federation — Delivery Lookup DGS Subgraph

- Designed and integrated a Netflix DGS subgraph for the Delivery Lookup service into CarGurus' federated GraphQL gateway, exposing vehicle delivery details data to frontend, backend, and AI consumers from a single graph.
- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies to avoid JIT compilation latency spikes on cold starts.
- Built load tests and staging GraphQL integration tests; migrating from legacy REST endpoints (2 consumers) to support the new AI chatbot/shopping assistant platform.

High-Traffic Legacy API Modernization

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core API behind CarGurus' most-visited page (~8,000 rpm, millions of daily impressions); the new implementation handled 50% of traffic during A/B testing.
- Introduced a new orchestration service and GraphQL boundary, consolidating 12 downstream integrations into 5 well-defined services, reducing fan-out and simplifying failure modes.
- Produced initial architecture in ~2 weeks; shipped the first production milestone in 6 months and completed the overall modernization ahead of schedule.

  p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)
- Defined architectural standards for new microservices; built a parity-testing framework validating full payload and data-integrity against the legacy stack; removed years of hidden domain logic with clear documentation.

ChatGPT Apps SDK MCP Server (0→1 POC)

- Sole backend engineer designing and implementing an MCP server compatible with the OpenAI Apps SDK, exposing CarGurus capabilities into ChatGPT as a high-visibility proof-of-concept.
- Delivered a performant implementation to unblock demos for PMs and directors, iterating on ambiguous requirements; designed for safe, incremental production rollout alongside existing APIs and clients.

Digital Shopping & Online Leads Products

- Led backend design and implementation for a new digital shopping product launched during the pandemic to capture online leads, using a microservice and event-driven architecture for inventory, merchandising, and lead flows.
- Subsequently led backend work for additional digital commerce experiments, enabling the company to quickly launch and safely retire product bets without destabilizing core systems.

Cross-Org Platform & API Leadership

- Established patterns for API design, caching, and distributed orchestration adopted across multiple teams; authored key design docs, RFCs, and decision records that drove consensus across platform and product teams.
- Mentored multiple senior engineers on system design, performance work, and path to Principal-level scope.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Authentication & Identity Modernization

- Upgraded legacy authentication to a modern Keycloak-based SSO platform, unifying auth across internal systems, an enterprise partner platform, and a large external community. Introduced social login and mobile login; migrated multiple Angular, Node.js, and Java monoliths behind the new auth gateway with minimal downtime.

Event Sourcing & CQRS Initiative

- Drove early design and prototyping of a microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.

Cross-Org Collaboration

- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations with clear observability and rollback paths.

Toast Inc. — Senior Software Engineer

Sep 2014 – Oct 2017

Monolith → Microservices Decomposition

- Among the earliest engineers during hypergrowth (first customer to hundreds in a few years); designed and implemented core backend services (Order Service, Labor Service, foundational APIs) that became the basis for future growth.
- Built initial AWS deployment pipelines before a dedicated DevOps team existed; created the first automated testing framework before dedicated QA, enabling safer releases during rapid product expansion.

Third-Party Integrations & Cross-Functional Leadership

- Led integrations with key partners (e.g., workforce management), ensuring reliable bidirectional sync and schema evolution. Partnered with PMs, sales, and support to translate early customer feedback into platform capabilities.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010